DSD Metrics Evidence Paper

What is ideal, acceptable and not acceptable for each smile-design metric, with the numbers pulled from dental papers

Scope and method. Searched PubMed for each metric the DSD software will measure and extracted the numbers from the abstracts of 39 papers (reference list at the end, with PMIDs and DOIs). **I read abstracts, not full texts.** Green / amber / red bands are my translation of the reported thresholds into software rules. Where I extrapolated beyond what a study tested, the cell says **(derived)**. Treat every band as a configurable default that you, the clinician, can override.

Bottom line

- **Strongest evidence (several concordant studies or systematic reviews):** gingival display, dental midline shift, diastema, occlusal plane cant. Build hard flags on these.
- **Weak evidence:** golden proportion, RED proportion and golden percentage are **not** found in natural dentition across populations (meta-analysis of 26 studies, 3,786 people, 12 countries [22]; golden percentage 25/15/10 not found [23]). Do not hard-code them. Offer them as optional presets only.
- **Central incisor width/length:** the preferred value in the one perception study that compared professionals and laypeople was 85%, not 75-80% [19]. My first guide quoted the older convention; Section 5 corrects it.
- **Professionals are stricter than laypeople** on almost every parameter [11][12][28]. Ship two threshold profiles: clinician (tight) and patient-facing (lay).
- **Gummy-smile tolerance differs by sex:** up to 3 mm for female and 1.5 mm for male smiles in a dynamic video study [3].
- **What this is not:** every threshold here comes from raters scoring digitally altered photos (often one model). It is perception data, not clinical outcome data.

Band colour key

Band	Meaning in the software
Ideal / target	The value raters scored highest, or the value the literature calls ideal. Display as the goal, never as a pass mark.
GREEN	Within the range that studies found equal to, or nearly equal to, the ideal. No action.
AMBER	Detectable to professionals and some laypeople. Show a review prompt to the clinician.
RED	Rated clearly unattractive, or beyond the tested range. Show a warning; the clinician decides.
Conf.	High = systematic reviews or several concordant studies. Moderate = 2+ concordant studies. Low = single study, convention, or conflicting data.

1. Smile-frame and soft-tissue metrics

Metric (unit)	Ideal / target	GREEN: accept	AMBER: review	RED: flag	Evidence (numbers from the cited papers; refs in brackets)	Conf.
Dental midline shift vs facial midline (mm)	0	0-2	More than 2, up to 3	More than 3	Laypeople accept up to 3 mm, with or without chin and nose in view [9]. Scores stayed near the unaltered image at 1 mm and 2 mm (88.9 and 86.8 vs 90.4) and fell to 57.5 at 4 mm [10]. Lay pooled threshold 0-3 mm [1]. Specialists detect smaller shifts than general dentists, who detect smaller than laypeople [11][12]. Medium effect size [16].	High
Midline cant (degrees)	0	0-3.5	3.5-5 (derived)	More than 5 (derived)	Midlines with 0-3.5 degrees of cant were among the most appealing; tested up to 6.5 degrees [11]. Medium effect size [16]. Cut-offs above 3.5 are my extrapolation.	Moderate
Incisal edge / occlusal plane cant vs interpupillary line (degrees)	0	0-2	2-4	More than 4	Scores dropped at 4 and 6 degrees (tested 0, 2, 4, 6) [14]. Lay threshold range 0-4 degrees [1]. Each 1 mm of cant lowered the rating by 3.6% [15]. Medium effect size [16]. Cant and gum show interact: the penalty for cant shrinks as gum show grows [7].	Moderate
Commissural line vs interpupillary line (degrees)	0	0-1	1-3 (derived)	More than 3 (derived)	Celebrity best smiles averaged 0.97 degrees vs 1.25 in dental students [34]. Small effect size [16]. When the commissure line is canted 3 degrees, 59-61% of laypeople preferred an occlusal plane that follows it [17]; prosthodontists and general dentists preferred a horizontal plane when the interpupillary line was canted [18]. Let the clinician choose the reference line.	Low-Mod.
Gingival display at full smile (mm; + = gum shown, - = tooth covered)	0 to -2	-2 to +2	+2 to +3 (male: +1.5 to +3)	More than +3	Pooled lay acceptance range +3 to -2.7 mm, median 0 mm (34 studies) [2]. Video study: acceptable to 3 mm in female and 1.5 mm in male smiles, steep drop after, rated very low above 6 mm [3]. 'Ugly' threshold 3 mm for dentists and 4 mm for non-professionals [4]. 0 mm preferred; 3 mm and 4 mm least attractive [5][6]. Orthodontists preferred 1 mm of lip coverage, dentists and laypeople 2 mm [7]. Systematic review range 1.5-4 mm [1].	High
Incisor display at rest (mm)	2-4 (young adult)	2-4	Not set	Not set	A review states 2-4 mm of maxillary incisor edge at rest as reasonable [8]. Age and lip length dependent. No threshold study retrieved, so no amber or red band.	Low
Smile arc	Consonant	Consonant	Flat	Reverse (inverted)	A consonant arc raised the mean rating by 12.6% over inconsonant [15]. Consonant rated most attractive [5][32][33]. Laypeople sometimes rate flat as equal or better [30][31]. Inverted arc had the strongest negative effect [32]. Small effect size [16].	Moderate
Buccal corridor (% of smile width)	10-15	10-15	5, or 20-25	Above 25 (untested)	At 15% raters gave the highest scores; 5% and 25% drew the most visual attention [14]. Conflicting: no corridor rated best in [5]; 38-48% of raters found wider corridors more attractive [37]; lay range in the systematic review is reported as 5-16 [1]. Culture dependent [36]. Weight this metric low.	Low
Visible teeth in the smile (count)	Report only	-	-	-	Celebrity best smiles showed 12 teeth in 22.2% of cases vs 6.2% of students [34]. Visible-tooth count affected social and psychological impact scores in a Han Chinese sample [39]. No threshold established.	Low

2. Tooth and gingival metrics

Metric (unit)	Ideal / target	GREEN: accept	AMBER: review	RED: flag	Evidence (numbers from the cited papers; refs in brackets)	Conf.
Central incisor width/length ratio (%)	80-85	75-85	70-75 or 85-90 (derived)	Below 70 or above 90 (derived)	Dentists and laypeople rated 85% (centrals) with 80% (laterals and canines) most esthetic of the three variants tested [19]. Natural central incisors in a Saudi cast sample: 8.74 by 9.84 mm, which is 89% (my calculation), a squarish form [20]. Small effect in laypeople, medium in dentists [16]. Population dependent.	Low-Mod.
Left-right central incisor width difference (mm)	0	0-1 (derived)	1-2 (derived)	2 or more	A single central incisor 2 mm off scored 45.5 vs 90.4 for the control; 1 mm spread across the arch scored 87.8 [10]. Unilateral width or length change of a central was least attractive in all rater groups [37]. Natural teeth show minor left-right asymmetry [20].	Moderate
Left-right central incisor edge height difference (mm)	0	0-0.5	0.5-1 (derived)	More than 1 (derived)	A central 0.5 mm shorter was one of only two asymmetries rated esthetic [29]. Incisor edge symmetry: medium effect in laypeople, large in the dental group [16].	Moderate
Left-right gingival margin difference, centrals (mm)	0	0-1	1-2	More than 2	Laypeople and preclinical students noticed only 2 mm or more; specialists, dentists and clinical students were stricter at every level [28]. A 2 mm gingival migration of one central was still rated esthetic [29]. Small effect size [16].	Moderate
Lateral vs central incisor position (mm)	Edge 1 shorter; margin 1 coronal	Edge 0-1 shorter; margin 0-1 coronal	Lateral edge level with or longer than central (derived)	-	All rater groups found a lateral gingival margin 1 mm coronal to the central, and a lateral edge 1 mm shorter, most attractive [27]. Equal margins scored 7.16 and a 1 mm difference 7.04 on a 10-point scale, so the difference is trivial [30].	Low-Mod.
Lateral incisor mesial angulation (degrees)	10-15	10-20	-	-	Dentists and laypeople scored 10 and 15 degrees highest; orthodontists scored 20 highest [27]. Other lateral changes (width, golden proportion, length) were not detected as unattractive [27].	Low
Diastema (mm)	0	0	Above 0, up to 2	More than 2	Lay threshold 0-2 mm [1]. Each 1 mm of diastema lowered the rating by 8.7% [15]. Large effect size in both lay and dental groups [16].	High
Central incisor crown length error (mm)	0	0-1 (derived)	1-2 (derived)	2 or more	The 'ugly' threshold for central incisor crown length change was 2.0 mm in both dentists and non-professionals [4].	Low-Mod.
Lateral / central apparent width (%)	63-70 (observed)	62-80	Not set	Not set	Observed: 63.7% in Saudi casts [20]; 67-70% in Kenyans [24]; RED ratio started near 69% in Gujarati students [25]. Golden 62% matched only 4.0% of Kenyans on one side [24]. An 80% ratio was rated more attractive than 62% and 70% [30].	Low
Canine / lateral apparent width (%)	78-84 (observed)	78-84	Not set	Not set	Observed: 78.4% in Saudi casts [20]; 82-84% in Kenyans [24]; up to 78.5% in Gujarati students [25].	Low
Golden percentage: central / lateral / canine (% of six-tooth width)	Report only	-	-	-	Classic 25 / 15 / 10 was not found in a systematic review of 16 studies [23]. Observed 22 / 15 / 12 in Kenyans [24] and 22 / 16 / 12 in Gujarati students [25]. Dentists preferred RED for normal-sized teeth and golden proportion for tall teeth [26].	Low

3. How much each metric matters: effect sizes

One study of 1,040 raters (laypeople, students, dentists, specialists) altered 13 parameters on one reference smile and measured how much each lowered attractiveness [16]. Use these as the starting weights in the scoring model. Negative values mean the alteration lowered the rating; larger magnitude means a bigger penalty. Classes are the authors': small 0.10-0.30, medium 0.30-0.50, large 0.50 and above.

Parameter	Laypeople	Dental group	Class (lay / dental)	Suggested weight
Color (shade)	-0.56	-0.56	Large / Large	High
Diastema	-0.55	-0.54	Large / Large	High
Incisor edge symmetry	-0.46	-0.54	Medium / Large	High
High smile (excess gum)	-0.42	-0.51	Medium / Large	High
Midline shift	-0.37	-0.49	Medium / Medium	Medium
Occlusal plane cant	-0.36	-0.49	Medium / Medium	Medium
Midline cant	-0.36	-0.48	Medium / Medium	Medium
Width-to-length tooth proportion	-0.26	-0.39	Small / Medium	Medium-low
Low smile (too little tooth)	-0.24	-0.23	Small / Small	Low
Commissure line cant	-0.15	-0.17	Small / Small	Low
Gingival margin level	-0.11	-0.01	Small / Small	Low
Teeth width proportion (golden-type)	-0.09	-0.10	Small / Small	Low
Inverted smile arch	-0.09	-0.21	Small / Small	Low

Implication for the AI: shade, diastema, edge symmetry and gum show matter far more than golden-type width proportions. A tool that spends its effort on proportion formulas and ignores these is optimizing the wrong thing. Note: [16] could not name a shared standard of beauty between laypeople and professionals, so keep weights adjustable.

4. Measured natural tooth dimensions (reference data)

Natural dimensions vary by population and by measurement method (casts vs clinical photos). Use these as population-specific priors, never as one universal norm.

Tooth	Mean width (mm)	Mean length (mm)	Width/length (my calc.)	Men: W / L (mm)	Women: W / L (mm)
Central incisor	8.74	9.84	88.8%	8.89 / 10.04	8.60 / 9.64
Lateral incisor	6.64	8.09	82.1%	6.79 / 8.30	6.49 / 7.89
Canine	7.82	9.08	86.1%	8.01 / 9.48	7.63 / 8.69

Source: Saudi sample, 180 adults aged 20-30, measured on stone casts with digital calipers [20]. A UK-resident Indian-origin sample (n = 100) found significant sex differences in tooth widths and no golden proportion relationships, and the authors advise against a formulaic left-right symmetrical approach [21]. The meta-analysis of 12 countries found no common proportion standard and recommends population-specific ratios [22].

5. Corrections to the first developer guide

Item in the first guide	What the literature in this paper shows	Action
Central incisor W/L 'about 75-80%'	Preferred value was 85% for centrals and 80% for laterals and canines [19]; natural Saudi cast mean is about 89% [20]. Classic 75-80% is a convention, not what these studies found.	Make the target a configurable range of 75-85%, default 80-85%.
Golden proportion and RED shown as design targets	Golden proportion matched 4.0% and 2.8% of Kenyans on one side [24]; meta-analysis found no consistent values [22].	Offer as optional presets. Never as pass/fail.
Gummy smile 'over about 3 mm excessive'	Consistent for the pooled lay range (+3 mm) [2], but male threshold was 1.5 mm in a dynamic study [3].	Use sex-specific amber thresholds.
Midline shift 'under about 2 mm tolerated'	Laypeople accept up to 3 mm [9]; professionals detect smaller shifts [11][12].	Green to 2 mm, amber to 3 mm; show clinician profile.
Incisor display at rest '1-3 mm in younger adults'	A review gives 2-4 mm [8]. Neither is backed by a threshold study.	Show as a report-only value with an age note.
Central crown length '10-11 mm'	Saudi cast mean was 9.84 mm [20]; casts measure anatomic crown, not clinical crown.	Treat length as population and method dependent.
Buccal corridor 'small looks fuller'	Evidence is conflicting [5][14][37] and culture dependent [36].	Low weight; no red flag.

6. Recommended software behavior

- Show each measured value next to its band and the reference number, for example: 'Gum display +3.5 mm, red, lay acceptance range to +3 mm [2]'.
- Two profiles: **Clinician** (professional-strict thresholds) and **Patient-facing** (lay thresholds). Default to clinician for internal planning.
- Sex-specific gingival display bands; population-specific proportion presets (cast-measured or photo-measured), selectable by the clinician.
- Weight the composite smile score by the effect sizes in Section 3. Do not average all metrics equally.
- Never present a band as a diagnosis. Bands are perception defaults; the clinician overrides and the override is logged.
- Record the unit and measurement method for every value (photo vs cast vs scan), because the studies disagree partly because of it.

7. Limitations you should weigh before relying on these numbers

- I extracted numbers from **abstracts only**. A threshold in an abstract may carry caveats in the full text. Before you publish or build a hard rule, open the paper behind that number.
- Most perception studies alter **one model's photo** and use small rater groups (for example 16 per group in [27]). Results may not generalize across faces, ages or sexes.
- Systematic review [1] reports heterogeneous data and could not run a meta-analysis; its buccal corridor values are given in units that differ from the primary studies (most use % of smile width).
- Culture changes thresholds: American laypeople were more tolerant than Iranian laypeople on diastema, gum display and buccal corridor [36]. Calibrate locally; an Egyptian-population rating study would be the best validation for your own tool.
- Classic foundational perception papers were not retrieved in this pass. If you want specific classic thresholds verified from source, that needs a follow-up search.
- Amber and red bands marked (derived) are my extrapolation and have no direct study behind them.

References (all indexed in PubMed; PMID and DOI)

[1] Laypeople's perceptions of frontal smile esthetics: a systematic review (66 studies). **PMID 27871700**; DOI 10.1016/j.ajodo.2016.06.022

[2] The threshold of acceptability of excessive gingival display by laypersons: a systematic review (34 studies). **PMID 35587001**

[3] Influence of maxillary gingival display on perception of smile aesthetics: dynamic smile visualisation study. **PMID 41604296**; DOI 10.1177/14653125251412487

[4] Perceptions of dentists and non-professionals on dental factors affecting smile aesthetics: Vietnam. **PMID 32138374**; DOI 10.3390/ijerph17051638

[5] Perception of altered smile esthetics including buccal corridor, gingival display, and smile arc by dentists and laypersons. **PMID 41846774**; DOI 10.4103/jpbs.jpbs_551_25

[6] Perception of smile attractiveness based on upper dental midline shift and gingival display. **PMID 41536395**; DOI 10.7759/cureus.99173

[7] The impact of occlusal plane cant along with gingival display on smile attractiveness. **PMID 26856663**; DOI 10.1111/ocr.12118

[8] Functional Aesthetic Occlusal Plane (FAOP). **PMID 27653271**; DOI 10.1590/2177-6709.21.4.114-125.sar

[9] The impact of midline deviation on smile attractiveness: cross-sectional study among laypersons. **PMID 40134997**; DOI 10.7759/cureus.79479

[10] Effect of midline deviation and crown width disproportion on perception of smile esthetics. **PMID 40405774**; DOI 10.1111/jerd.13492

[11] Attractive levels of dental midline shift, midline cant, mouth corner, nose and chin asymmetry; perception thresholds. **PMID 42851635**; DOI 10.1002/hsr2.73348

[12] Comparative assessment of smile attractiveness perception and treatment need across dental professionals and laypersons. **PMID 41209948**; DOI 10.7759/cureus.94037

[13] Perception of discrepancy in upper midline position with gingival display, various occupations in Iran. **PMID 36967115**; DOI 10.4041/kjod22.164

[14] The impact of dentolabial variables on esthetic perception: an eye tracking study. **PMID 41776802**; DOI 10.4103/njcp.njcp_42_25

[15] Comparison of pleasant and unpleasant smile characteristics in the perception of laypeople, Iranian population. **PMID 37565945**; DOI 10.1016/j.ajodo.2023.04.022

[16] Analysis of the effect magnitude of different parameters on smile attractiveness. **PMID 34796640**; DOI 10.1111/jerd.12834

[17] Layperson's preference regarding orientation of the transverse occlusal plane and commissure line. **PMID 27881322**; DOI 10.1016/j.prosdent.2016.07.027

[18] Preferences of laypeople, general dentists and specialists on occlusal plane and dental midline in asymmetrical faces. **PMID 40771656**; DOI 10.1155/ijod/4232915

[19] Width/length ratio in maxillary anterior teeth: esthetic preferences among professionals and laypersons. **PMID 28513973**; DOI 10.1111/jopr.12642

[20] Maxillary anterior teeth dimension and relative width proportion in a Saudi subpopulation. **PMID 33897325**; DOI 10.1016/j.jtumed.2020.12.009

[21] Upper anterior tooth dimensions in a young-adult Indian population in the UK. **PMID 29171516**; DOI 10.1038/sj.bdj.2017.986

[22] Parameters related to the golden ratio in teeth of individuals from different countries: systematic review and meta-analysis. **PMID 37932181**; DOI 10.1016/j.prosdent.2023.09.032

[23] Evaluation of golden percentage in natural maxillary anterior teeth width: a systematic review. **PMID 34340826**; DOI 10.1016/j.prosdent.2021.06.015

[24] Golden proportion, golden percentage and RED proportion in Kenyans of African descent. **PMID 38970240**; DOI 10.1002/cre2.923

[25] Prevalence of golden percentage, RED and Tarulatha-Ruchi proportion in dental students. **PMID 39926964**; DOI 10.4103/jpbs.jpbs_845_24

[26] Smile aesthetics in Pakistani population: dentist preferences on anterior teeth proportion and harmony. **PMID 38553673**; DOI 10.1186/s12903-024-04100-4

[27] Impact of variations in maxillary lateral incisor dimensions on smile esthetics. **PMID 34417035**; DOI 10.1016/j.ajodo.2020.06.047

[28] Impact of gingival margin asymmetries on smile esthetic perception. **PMID 35821722**; DOI 10.15644/asc56/2/7

[29] Esthetic evaluation of dental and gingival asymmetries. **PMID 25986710**; DOI 10.1016/j.ortho.2015.03.005

[30] Perception of altered smile esthetics by orthodontists, general dentists and laypeople (gingival margin, golden ratio, smile arc). **PMID 42038189**; DOI 10.31661/gmj.vi.3947

[31] Influence of smile arc on esthetic perception in relation to gingival display: Iranian population. **PMID 42146272**; DOI 10.18502/fid.v22i46.20328

[32] Impact of facial pattern and smile arc on perception of male facial attractiveness. **PMID 42908359**; DOI 10.1111/jerd.70264

[33] Evaluation of aesthetics of posed smiles based on smile-related characteristics. **PMID 40119218**; DOI 10.1007/s10006-025-01364-0

[34] Quantification of facial and smile esthetics. **PMID 28552284**; DOI 10.1016/j.prosdent.2017.04.002

[35] Perception of smile attractiveness regarding the buccal corridor space as defined by the eyes. **PMID 36628925**; DOI 10.1111/jerd.13005

[36] Laypersons' perception of smile esthetics from different backgrounds. **PMID 39188396**

[37] Perception of smile attractiveness among orthodontists, restorative dentists and laypersons in Saudi Arabia. **PMID 30820086**; DOI 10.4103/JCD.JCD_429_18

[38] Evaluating golden and RED proportions using two methods in an urban Tirupati population. **PMID 42695117**; DOI 10.7759/cureus.113953

[39] Effect of smile esthetics on quality of life in a Han Chinese population. **PMID 36573304**; DOI 10.1111/jerd.12999